

Abstract:

Among the most prevalent brain tumors, Glioblastoma Multiforme (GBM) is the most malignant tumor. Currently, the tumor is treated with various traditional methods. The usage of epigenetic inhibitors that has a crucial role in the histone methylation process, possess the ability to inhibit the tumor suppressor genes in most of the cancer tissues. It has been a matter of great interest for researchers in treating Glioblastomas. This approach was taken as Chemotherapies weren't fruitful. The project focuses on three epigenetic EZH2 inhibitor drugs. EZH2 inhibitors focus on histone methylation that causes transcriptional repression. Based on the prior research conducted by Mishra and team, (Mishra et al., 2020) where EPZ 6438 (EZH2 inhibitor) was utilized to treat GBM, the three components (EPZ 3468, UNC 1999 and GSK 343) were selected for analysis. In the previous in vitro analysis, the components had proved to inhibit cell proliferation and cause cell loss in GBM cell lines. As nearly 80-90% cells were lost as days passed. The project focuses on finding the probable cause of loss of cells by identifying the significant genes. The concluding hypothesis of the project is that "significant genes in the components have biological functions either to differentiate as neural cell types or to regulate the protein synthesis pathways to cause the cell loss in GBM cell lines".

Keywords: Methylation, Inhibition, Cell proliferation, Cell differentiation.